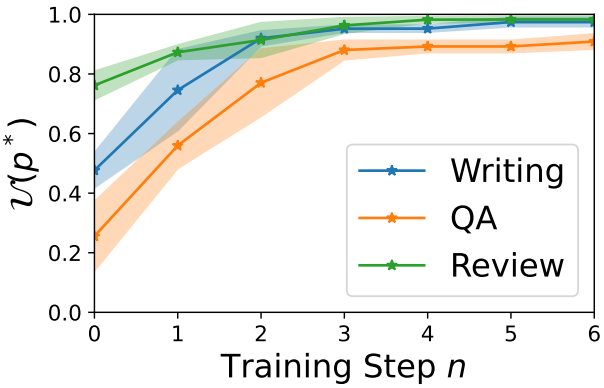

Figure 3: The trajectory of the $\mathcal{U}(p^*)$ during prompt optimization. This plot is derived from three distinct training runs on three tasks.

GPT-4-0613 OpenAI (2022). We employ Chat-D as proxy detector. The results in Table 13 indicate that SICO with different LLMs maintains a high degree of detection evasion performance.

Table 13: The AUC scores of SICO using different LLMs. "-" indicates the performance without applying any evasion method. Symbol "*" represents that SICO uses GPT3-D as the proxy detector for prompt construction.

Dataset	LLM	GPT3-D*	GPT2-D	DetectGPT	Log-Rank
Writing	–	0.908	0.848	0.834	0.914
	WizardLM	0.571	0.582	0.510	0.481
	GPT-3.5	0.239	0.332	0.149	0.147
	GPT-4	0.157	0.260	0.263	0.277
QA	–	0.981	0.906	0.876	0.956
	WizardLM	0.721	0.467	0.561	0.404
	GPT-3.5	0.402	0.566	0.178	0.183
	GPT-4	0.414	0.369	0.344	0.287
Review	–	0.925	0.952	0.808	0.982
	WizardLM	0.707	0.402	0.570	0.542
	GPT-3.5	0.465	0.264	0.270	0.300
	GPT-4	0.408	0.307	0.427	0.382

C Detectors

In this section, we introduce the mechanism and settings of the detectors in our experiments.

C.1 GPT-3.5 Detector

GPT-3.5 detector is trained on Human ChatGPT Comparison Corpus (HC3) dataset (Guo et al., 2023), which including answers generated by ChatGPT and human. English-version of HC3 dataset contains five splits: reddit-eli5, OpenQA, Wikipedia, medicine and finance. The base model is RoBERTa-base and we use the model that only take answers as input.

C.2 GPT2 Detector

GPT-2 detector is obtained by fine-tuning a RoBERTa model with the outputs of the 1.5B-parameter GPT-2 model. The detector and the GPT-2 output dataset are both provided by OpenAI. Although it is trained on GPT-2 outputs, our experiments shows that it can effectively detect text from GPT-3.5.

C.3 DetectGPT

DetectGPT identifies if a text is generated by a model by observing a unique characteristic: AI-generated text tends to be in areas where the language model's log probability function has a negative curve. Here's how it works: It first perturbs the input text and constructs multiple perturbations of input text. The perturb step is completed by a mask language model. Then it checks the log probability of these variations against the original text by a inner language model. Finally, the text is considered AI-generated if the log probability of the original input text is significantly higher than the log probability of perturbations.

We use z-score implementation of DetectGPT and set sample number to 100 and replacement ratio to 0.3. The inner language model is GPT2-medium and the mask language model is t5-large (Raffel et al., 2020).

C.4 Log-Rank

Log-Rank method employs the mean prediction rank of each token in a text. Specifically, for each word in a text, given its previous context, it can calculate the absolute rank of this word by an inner language model. Then, for a given text, we compute the score of the text by averaging the rank value of each word. Note that a smaller score denotes the text is more likely to be machine-generated. In experiment, we use GPT2-medium to calculate the rank of tokens to align with the implementation of DetectGPT.

C.5 GPTzero

GPTzero is a recently proposed commercial detector, employed by many users and oragnizations. As claimed in its websites[5], GPTzero can be used to detect the outputs from detect ChatGPT, GPT4, Bard, LLaMa, and other AI models. GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. GPTzero was trained on a large, diverse corpus of human-written and AI-generated text, with a focus on English prose. GPTZero has served over 2.5 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

C.6 OpenAI Detector

OpenAI detector is officially provided by OpenAI after the release of ChatGPT. Although it only offers a web interface, we adopted the API implementation from (Krishna et al., 2023), which uses "model-detect-v2" in the OpenAI API. Through reverse engineering of the website, we determined that the web interfaces indeed call this model.

On July 20, 2023, OpenAI discontinued this detector, "due to its low rate of accuracy."[6]. Considering the discontinuation of the OpenAI detector aligns with our findings, we choose to present the results of it in our paper, though it is out of date.

D Human Evaluation Details

For each text, we show two questions for the human annotator. In terms of readability, we present the human annotator five options, with the scale of 1 to 5. The higher the value, the more readable of the presented texts. The question is identical for three tasks. The actual question and options are presented in Table 14. For task completion rate, we design three task-specific questions, as show in Table 15. Figure 5 shows the

[5]https://gptzero.me/
[6]https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text

interface of our annotation platform. We estimated that the evaluation time of each text ranges from 60 to 90 seconds. For the imperceptibility experiment, we present the annotators with text and a question to determine if ChatGPT generates the text. All annotators in the experiment used ChatGPT before.

Table 14: Question and options designed for readability.

Question	On a scale of 1-5, how would you rate the readability of the given text?
Options	1: Very difficult to read and understand. 2: Difficult to read, need extra time to understand. 3: Neutral. 4: Easy to read. 5: Very clear and easy to read.

Table 15: Question and options designed for task completion.

	Writing
Question	Is this academic essay correctly written (No errors and in academic style)?
Options	No, it has some obvious errors or not in academic format.
	Yes, it is correctly written.
	QA
Question	Does the answer relate to the question? (regardless of correctness)
Options	No, not relative.
	Yes, it is relative.
	Review
Question	Does this review impact your decision to choose this service?
Options	No influence.
	Yes, it provides useful information.

Human evaluation on Parrot. As Parrot method performs better than DIPPER in the Writing task, we conducted a small experiment to evaluate the usability of text generated text, similar to Table 2. We randomly sampled 120 examples (40 for Parrot, SICO-Para and SICO-Gen) from the writing task and asked two human annotators to evaluate them. The experiment result in Table 16 shows that SICO still outperforms Parrot by a large margin.

Table 16: Human evaluation results of Parrot and SICO.

Method	Readablity	Task Compeletion Rate
Parrot	3.35	70.0
SICO-Para	3.80	82.5
SICO-Gen	**3.90**	**85.0**

E Datasets

Table 17 presents the prompts we employed to create the initial AI-generated text y_{AI}. For academic writing, we sample Wikipedia paragraphs from SQuAD dataset. Then we give GPT-3.5 the first 30 words of these paragraphs and ask GPT-3.5 to complete the rest.